FIITJEE

ALL INDIA TEST SERIES

PART TEST – III

JEE (Main)-2020

TEST DATE: 08-12-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

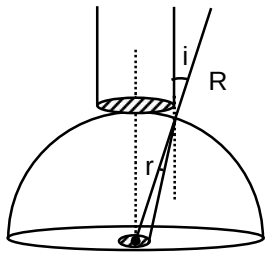
1. D

Sol. Using Snell's law:

$\sin i = \frac{R}{2R} \Rightarrow i = 30°$

$\sin i = \mu \sin r$

$\sin r = \frac{1}{2\sqrt{2}}$



2. C

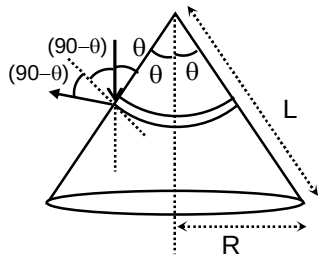
Sol. $\frac{10 - 0.7}{2 \times 10^{-3}} = 4.65\,k\Omega$

3. D

Sol. $F = \int \frac{2Ids\sin^2\theta}{c}\sin\theta$

$= = \frac{2I\sin^2\theta}{c}\sin\theta\int ds = \frac{2I\sin^2\theta}{c}\pi RL\sin\theta$

$= \frac{I\pi R^2}{c}(1 - \cos 2\theta)$



4. A

Sol. $y(x, t = t_0) = A \cos kx$

$\therefore\ y(x, t) = A \cos k[x + v(t - t_0)]$

$y(x, t) = A \cos [kx + \omega t - \omega t_0)]$

at $t = 2t_0$

$y(x, t = 2t_0) = A \cos [kx + \omega t_0)]$

$y(x, t = 2t_0) = A \cos [kx + 2\pi f t_0]$

5. A

Sol. $A \xrightarrow{\lambda_1} B \xrightarrow{\lambda_2} C$

$$\frac{dN_B}{dt} = \lambda_1 N_A - \lambda_2 N_B = 0$$

$$\Rightarrow \frac{N_A}{N_B} = \frac{\lambda_2}{\lambda_1} \Rightarrow \frac{\lambda_1 N_A}{\lambda_2 N_B} = 1$$

6. A

Sol. impulse momentum theorem

$\rho v v_1 A\Delta t = \Delta p A \Delta t$

$\Rightarrow \Delta p = \rho v v_1$

Also $\Delta p.(2r)\ell = \sigma.(2t)\ell$

$\Rightarrow \Delta p r = \sigma t$

$\Rightarrow \rho v v_1 r = \sigma t$

$$v_1 = \frac{\sigma t}{\rho v r}$$

7. B

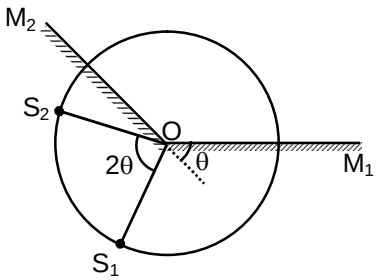
Sol. $|Q| = 0.00394 \times 931.5 = 3.67$ MeV

threshold Kinetic energy

$$= \left(\frac{m_1 + m_2}{m_2}\right)|Q| = \left(1 + \frac{m_P}{m_N}\right)|Q| = 3.91\,\text{MeV}$$

8. B

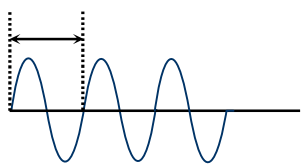
Sol. If mirror rotates by angle θ image of object rotates by angle 2θ.



9. B

Sol. Optical path = μ geometrical path

$$\frac{\left(\frac{3\lambda}{\mu_1}\right)}{\left(\frac{2\lambda}{\mu_2}\right)} = \frac{1}{2} \Rightarrow \frac{3\mu_2}{2\mu_1} = \frac{1}{2} \Rightarrow \mu_1 = 9/2$$



10. A

Sol. $\lambda = \dfrac{h}{\sqrt{2mK}}$

$\lambda = \sqrt{\dfrac{150}{75}} \Rightarrow \lambda = 1.414\ \text{Å}$

For the first maxima

$2d\sin\theta = \lambda$

$\Rightarrow \sin\theta = \dfrac{\lambda}{2d} = \dfrac{1.414}{2\times 30} = 0.023$

11. B

Sol. The direction of $(\vec{E}\times\vec{B})$ gives the direction of propagation of wave

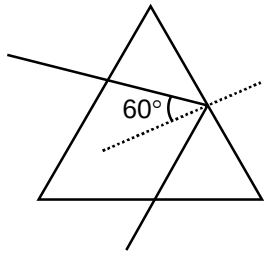
$\hat{i}\times\hat{j}\rightarrow\hat{k}$

12. A

Sol. $\dfrac{\omega^2 xT}{x} = \omega^2 T$

13. C

Sol. TIR will take place



14. C

Sol. $\dfrac{3}{2v_1} - \dfrac{1}{\infty} = \dfrac{(0.5)}{-20}$

$\dfrac{2}{f} - \dfrac{3}{2v_1} = \dfrac{0.5}{-30}$

$\dfrac{2}{f} = -\dfrac{1}{2}\left(\dfrac{1}{20} + \dfrac{1}{30}\right)$

$f = -\dfrac{4\times 20\times 30}{50} = -48\ \text{cm}$

15. B

Sol. Total angular dispersion = $(\mu_v - \mu_r)A - (\mu'_v - \mu'_r)A' = 0$

16. C

Sol. In equilibrium

$\sigma a^3 g = \rho a^2 x_0 g$

$x_0 = \dfrac{\sigma a}{\rho}$

The work done by the applied force,

$$\Delta W = \int F dx = \int_0^{a-x_0} \rho a^2 x g dx$$

$$= \frac{\rho a^2 g}{2}(a - x_0)^2 = \frac{\rho a^2 g}{2}\left[a - \left(\frac{\sigma a}{\rho}\right)\right]^2$$

$$= a^4 g\left(\frac{\rho}{2} - \sigma + \frac{\sigma^2}{2\rho}\right)$$

17. A

Sol. Using conservation of energy, $K - \frac{GMm}{r} = 0$

$$K = \frac{GMm}{r} \Rightarrow K = (GMm)\frac{1}{r}$$

$$GMm = \frac{3}{2} \times 10^{18}$$

$$m = \frac{3}{2MG} \times 10^{18} Kg$$

18. B

Sol. $t = \frac{T}{2\pi}(2\pi - 2\phi) = \frac{T}{\pi}(\pi - \phi)$

$$\phi = \tan^{-1}\left(\frac{vT}{2\pi x}\right) = \tan^{-1}\left(\frac{4\pi}{2\pi \cdot 2}\right)$$

$\phi = \pi/4$

$$t = 4 \times \frac{3\pi}{4} = 3\pi \sec$$

19. C

Sol. $\Delta P = \frac{2T}{r}$

$$\therefore\ P = P_0 + \rho g H + \frac{2T}{r}$$

20. D

Sol. Bernoulli's theorem

SECTION – B

21. 1

Sol. $ma = -\left(\frac{m}{\ell} \cdot x\right) g \sin\theta$

$$a = -\left(\frac{g \sin\theta}{\ell}\right) x$$

$$\Rightarrow \quad \frac{d^2x}{dt^2} = -\left(\frac{g \sin\theta}{\ell}\right) x$$

$$T = 2\pi\sqrt{\frac{\ell}{g\sin\theta}} = 2\pi\sqrt{\frac{125}{10\times\frac{1}{2}}} = 2\pi\times 5 = 10\pi \text{ sec}$$

Time spent on the inclined track is $t_0 = \frac{T}{2} = \frac{10\pi}{2} = 5\pi$ sec

22. 4

Sol. $L^2 = [M^{-1}L^3T^{-2}]^{\alpha}\,[LT^{-1}]^{\beta}\,[M]^{\gamma}$

$-\alpha + \gamma = 0$

$-2\alpha - \beta = 0$

$3\alpha + \beta = 2$

$\Rightarrow \alpha = 2, \beta = -4, \gamma = 2$

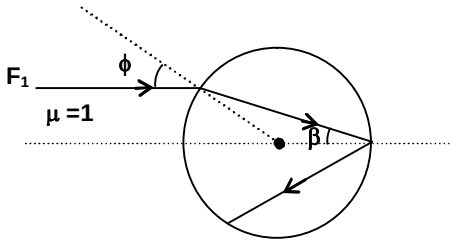
SECTION – C

23. 00062.50

Sol. $\sin\phi = \mu \sin\beta$

$\cos\beta = \mu/2 = 5/8$ (since $\phi/2 = \beta$)

$\therefore\ 100\cos\beta = 100\times\frac{5}{8} = 62.50$



24. 00015.00

Sol. $\Delta p = \frac{mg}{A} - \frac{mg}{4A}$

$\Rightarrow \qquad \rho a\ell = \frac{3mg}{4A}$

$$a = \frac{3mg}{4A\rho\ell} = \frac{3\times 1\times 10}{4\times 2.5\times 10^{-4}\times 2\times 10^{3}\times 1} = 15 \text{ m/s}^2$$

25. 00000.02

Sol. $\left|\frac{\Delta V}{V}\right| = \frac{\Delta P}{B} = \frac{99\times 10^5}{4.95\times 10^8} = 0.02$

Chemistry

PART – II

SECTION – A

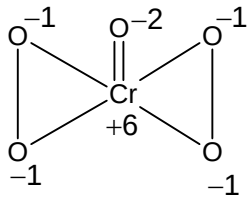
26. B

Sol. Balanced reaction is

$$2CrO_5 + 3H_2SO_4 \longrightarrow Cr_2(SO_4)_3 + 3H_2O + \frac{7}{2}O_2$$

So, 1 mole CrO_5 liberates $\frac{7}{4}$ moles of O_2.

It is a redox reaction because oxidation number of Cr decreases and oxidation number of O increases.



27. A

Sol. Initial weight of $H_2O(\ell) = 18$ g

$$n_{H_2O}(g) = \frac{PV}{RT}$$

$$= \frac{\left(\frac{24.63}{760}\right) \times 7.6}{0.0821 \times 300} = 0.01 \text{ mole}$$

% wt. of H_2O vapourised $= \frac{0.01 \times 18}{18} \times 100$

= 1%

28. C

Sol. $\Lambda_m = \frac{\kappa \times 1000}{M}$

or $\Lambda_m = \frac{1}{\rho} \times \frac{1}{M} \times 1000$ [κ = specific conductance, ρ = specific resistance]

or $\Lambda_m = \frac{1000}{xy}$

29. A

Sol. NaCl is non-volatile but CH_3OH is volatile. So, these solutions would have different boiling points. Raoult's law can not be applied to CH_3OH solution since it does not form an ideal solution with H_2O.

30. D

Sol. Electrophoresis means movement of colloidal particles (particles of dispersed phase) under the influence of electric field.

31. D

Sol. (A) Stronger the inter-atomic interaction, higher the enthalpy of atomization.

(B) Cr^+ has half-filled valence shell electronic configuration ($3d^5$), so it has higher second ionization energy. Similarly, Cu^+ has completely filled valence shell electronic configuration $3d^{10}$, so it has higher second ionization energy.

(C) Sum of first two ionization energies ($IE_1 + IE_2$) of Ni(II) is less than that of Pt(II), while sum of first four ionization energies ($IE_1 + IE_2 + IE_3 + IE_4$) of Pt (IV) is less than that of Ni(IV).

(D) Occurs in the middle, e.g. Mn has +7 oxidation state.

32. D

Sol. White precipitates of Hg_2Cl_2 and AgCl are water insoluble (even in hot water). Copper salts do not give white precipitate with dilute HCl as $CuCl_2$ is water soluble (at room temperature)

$$Pb^{+2} + 2Cl^- \longrightarrow \underset{\left(\begin{array}{l}\text{white precipitate}\\ \text{soluble in hot water}\end{array}\right)}{PbCl_2}$$

$$Pb^{+2} + S^{-2} \longrightarrow \underset{(\text{Black})}{PbS}$$

33. D

Sol. $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$

34. B

Sol. $t_{1/2} \propto [A]_0^{1-n}$

$100 \propto (0.1)^{1-n}$

$50 \propto (0.025)^{1-n}$

On dividing

$2 = 4^{1-n}$

$2 = \left(2^2\right)^{1-n}$

$2^1 = 2^{2-2n}$

$\therefore 2 - 2n = 1$

or $2n = 1$

or $n = \frac{1}{2}$

$100 \propto (0.1)^{1/2}$

$t_{1/2} \propto (1)^{1/2}$

Divide

$$\frac{t_{1/2}}{100} = \left(\frac{1}{0.1}\right)^{\frac{1}{2}}$$

$\therefore t_{1/2} = 100\sqrt{10}$ minutes.

35. B

Sol. Siderite is an ore of iron ($FeCO_3$).

36. C

Sol. Colour is attributed to the presence of unpaired f-electrons.

$Ce^{+3} = 4f^1$, $Sm^{+3} = 4f^5$

$La^{+3} = 4f^0$, $Lu^{+3} = 4f^{14}$

37. A

Sol. $$Z = \frac{PV}{nRT} = \frac{PV}{\frac{w}{M} \times RT} = \frac{PM}{dRT}$$

$$Z = \frac{8.314 \times 10^4 \times 18 \times 10^{-3}}{0.8 \times 8.314 \times 300} = \frac{18 \times 10}{300 \times 0.8}$$

$$Z = \frac{3}{4} = 0.75$$

38. C

Sol. $$U = \sqrt{\frac{3RT}{M}}$$

Let new rms speed be U_2.

$$U_2 = \sqrt{\frac{3R \times 2T}{\frac{M}{2}}} = 2 \times \sqrt{\frac{3RT}{M}}$$

$$\therefore U_2 = 2U$$

39. B

Sol. $C_{P,m} = aT^3$

$$a = \frac{C_{P,m}}{T^3} = \frac{0.84}{(10)^3} = 0.84 \times 10^{-3}$$

$$S_m = \int_{0K}^{10K} \frac{C_{P,m}}{T}.dT$$

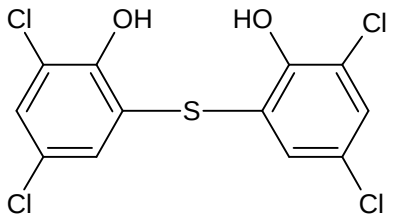
$$S_m = \int_{0K}^{10K} \frac{aT^3}{T}.dT = \int_{0K}^{10K} aT^2.dT$$

$$S_m = \frac{a}{3}\left[10^3 - 0\right] = \frac{a}{3} \times 10^3$$

$$S_m = \frac{0.84 \times 10^{-3} \times 10^3}{3} = 0.28$$

40. B

Sol.



Bithionol

There are 26 σ bonds and 6π bonds. So, ratio is $\frac{26}{6}=\frac{13}{3}$

41. C

Sol. In order to get $\left[H_3O^+\right]=1M$, H_2SO_4 concentration should be = 0.5 M.

$$\underset{(H_2SO_4)}{M}=\frac{W(\text{in g})}{\text{Molar mass}\times V\text{ in L}}$$

$$0.5=\frac{W(\text{in g})}{98\times 0.2}$$

Wt. of H_2SO_4 = 9.8 g

42. A

Sol. Cu^{+2} is member of II group cations while Ni^{+2}, Co^{+2} and Mn^{+2} are members of IV group cations. When H_2S is passed in presence of dilute HCl, only II group cation (Cu^{+2}) is precipitated.

43. A

Sol. Following reactions are responsible for the depletion of ozone layer

$$NO+O_3\longrightarrow NO_2+O_2$$

$$CFCl_3\xrightarrow{h\nu}\dot{C}F_2Cl+Cl^{\bullet}$$

$$Cl^{\bullet}+O_3\longrightarrow ClO^{\bullet}+O_2$$

$$ClO^{\bullet}+O\longrightarrow\dot{Cl}+O_2$$

CH_4 is not responsible for ozone layer depletion.

44. A

Sol. $\left[H^+\right]=10^{-2}$

m moles of $H^+=10^{-2}\times 100=1$

During electrolysis, following reactions take place.

At cathode

$$Cu^{+2}(aq.)+2e^-\longrightarrow Cu(s)$$

At anode

$$H_2O(\ell)\longrightarrow\frac{1}{2}O_2(g)+2H^+(aq.)+2e^-$$

From above reactions, it is clear that for 1 m mole of H^+ produced at anode, 0.5 m mole of Cu^{+2} is reduced at cathode.

Initial m mole of Cu^{+2} = 100 mL × 0.05

= 5 m mole

After electrolysis

Cu^{+2} remaining in solution = 5 – 0.5 = 4.5 m moles

$$2Cu^{+2}+4I^-\longrightarrow Cu_2I_2+I_2$$

So, according to above reaction

4.5 m mole Cu^{+2} will liberate $\frac{4.5}{2}$ m moles of I_2.

$$I_2+2Na_2S_2O_3\longrightarrow Na_2S_4O_6+2NaI$$

According to above reaction $\frac{4.5}{2}$ m moles of I_2 requires 4.5 m moles of $Na_2S_2O_3$.

$$M = \frac{\text{m moles}}{\text{V in mL}}$$

$$V(\text{in mL}) = \frac{4.5}{0.04} = \frac{450}{4}$$

V = 112.5 mL

45. C

Sol. Excess of KI is added, I^- ions are adsorbed on AgI forming a fixed layer and give – ve charge on colloid particles. It then attracts K^+ ions from the medium forming second diffused (mobile) layer.

SECTION – B

46. 8

Sol. Total 8 f-Block elements in 4f-series have odd electrons in f-orbitals in their ground state configuration.

$_{58}Ce, _{59}Pr, _{61}Pm, _{63}Eu, _{64}Gd, _{65}Tb, _{67}Ho, _{69}Tm$

47. 6

Sol. Sucrolose is 600 times sweet in comparison to cane sugar and Aspartame is 100 times sweet in comparison to cane sugar.

So, $\frac{x}{y} = \frac{600}{100}$

= 6

SECTION – C

48. 00036.85

Sol. $\log_{10} K = \log_{10} A - \frac{Ea}{2.303RT}$

$$\text{Slope} = -\frac{Ea}{2.303R}$$

$$-8000 = -\frac{Ea}{2.303 \times 2}$$

$$Ea = 8000 \times 2 \times 2.303 = 36\,848 \text{ calorie}$$

$$Ea = 36.85 \text{ Kcal}$$

49. 00018.60

Sol. $2Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$

n-factor of I_2 = 2

Therefore, number of m moles of $I_2 = 15 \times \frac{M}{40} = \frac{15}{40}$

Number of m moles of $Na_2S_2O_3 = \frac{15}{40} \times 2 = \frac{15}{20}$ in 10 mL solution.

Weight of $Na_2S_2O_3.5H_2O(\text{in g})$ in 10 mL solution $= \frac{15}{20} \times 10^{-3} \times 248 = 0.186$ g

Therefore, concentration of $Na_2S_2O_3.5H_2O$ = 18.6 (in g/L)

50. 00001.09

Sol. $\Delta G^o = -nFE^o_{cell}$

$$E^o_{cell} = \frac{-2746 \times 1000}{-26 \times 96500} = 1.09 \text{ V}$$

Mathematics

PART – III

SECTION – A

51. A

Sol. Vector $(\vec{v}) = \dfrac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|} = \dfrac{1}{\sqrt{35}}\left(5\hat{i}-3\hat{j}-\hat{k}\right)$

52. D

Sol. $(\vec{a}-\vec{b})\cdot\left[(\vec{b}-\vec{c})\times(\vec{c}-\vec{a})\right] = \left[\vec{a}\ \vec{b}\ \vec{c}\right]-\left[\vec{b}\ \vec{c}\ \vec{a}\right] = 0$

53. C

Sol. Required projection is $\dfrac{2}{7}(-1-1)+\dfrac{3}{7}(4-2)-\dfrac{6}{7}(2-3) = \dfrac{8}{7}$

54. C

Sol. D.R. of normal ≡ (2, 1, 1). Let, Q be the image of point P in the plane and PQ meet in R
R ≡ (2r + 3, r + 5, r + 7), ∴ 2(2r + 3) + r + 5 + r + 7 = 0, ∴ r = –3
∴ Using section formula Q ≡ (–9, –1, 1)

55. D

Sol. The number of terms is same as $(1 + x + x^2)^n$ which is 2n + 1

56. B

Sol. $\dfrac{1}{{}^{2n+1}C_r}+\dfrac{1}{{}^{2n+1}C_{r+1}} = \dfrac{{}^{2n+2}C_{r+1}}{{}^{2n+1}C_r\cdot{}^{2n+1}C_{r+1}} = \dfrac{2n+1}{2n+1}\times\dfrac{1}{{}^{2n}C_r}$

57. C

Sol. $\alpha = \lambda(1-\beta) = \lambda-\lambda(\lambda-\lambda c)$
Also, $(1+\lambda^3)\beta = \lambda-\lambda^2+\lambda^3$, $(1+\lambda^3)\gamma = \lambda-\lambda^2+\lambda^3$
∴ $(1+\lambda^3)\alpha = (1+\lambda^3)\beta = (1+\lambda^3)\gamma$, ∴ $\lambda^3 = -1$, ∴ $\lambda = -1, -\omega, -\omega^2$

58. B

Sol. Roots are 3 – i and –2 + i, ∴ product is –6

59. D

Sol. $p = \dfrac{3}{4}$, $q = \dfrac{1}{4}$, n = 5

Probability = ${}^5C_3\left(\dfrac{3}{4}\right)^3\left(\dfrac{1}{4}\right)^2+{}^5C_4\left(\dfrac{3}{4}\right)^4\left(\dfrac{1}{4}\right)+{}^5C_5\left(\dfrac{3}{4}\right)^5 = \dfrac{459}{512}$

60. A

Sol. $P(E\cap F) = P(E)\cdot P(F) = \dfrac{1}{12}$, $P(E'\cap F') = \dfrac{1}{2}$, $(1-P(E))(1-P(F)) = \dfrac{1}{2}$

61. D

Sol. $C_1 \to C_1 - C_2$, $C_2 \to C_2 - C_3$, open by R_1, ab + abc + ac + bc = 0
∴ $\dfrac{1}{a}+\dfrac{1}{b}+\dfrac{1}{c} = -1$

62. B

Sol. $\left|\text{Adj}\left(\lambda I_n\right)\right| = \lambda^{n(n-1)}$

63. B

Sol. Order of $B' \to 4 \times 3$, order of $B'A \to (4 \times 3) \times (3 \times 4) \Rightarrow 4 \times 4$

64. C

Sol. $(x + \alpha) = 1$ and $x + 1999 = -1 \Rightarrow \alpha = 1993$
$x + \alpha = -1$ and $x + 1991 = 1$, $\alpha = 1989$
Sum = 1993 + 1989

65. D

Sol. Let term be $a - 2d$, $a - d$, a, $a + d$, $a + 2d$
$\therefore\ 3a = -12 \Rightarrow a = -4$, $(a - 2a)(a - d)(a) = 8$
$\therefore\ d = -3$
$\therefore\ a_2 + a_4 + a_6 = -21$

66. C

Sol. Required value is $2 \times \underline{\lfloor 4} \times \dfrac{\underline{\lfloor 4}}{\underline{\lfloor 3}} = 192$

67. D

Sol. Number of ways is ${}^8C_2 \times (3)^6$

68. A

Sol. Let numbers be $a_1, a_2, \ldots\ldots, a_{10}$. $\sum_{i=1}^{10} a_i = 125$, $\sum_{i=1}^{6} a_i = 90$, $\sum_{i=6}^{10} a_i = 50$

$\therefore\ a_6 + 125 = 90 + 50\ \therefore\ a_6 = 15$

69. C

Sol. $\bar{x} = \dfrac{1+2+3+\ldots\ldots+2n+1}{2n+1} = n+1,\ \sigma^2 = \dfrac{1}{2n+1} \times \sum_{r=0}^{2n} \left((1+r)-(1+n)\right)^2 = \dfrac{n(n+1)}{3}$

$\therefore\ \sigma^2 = \sqrt{\dfrac{n(n+1)}{3}}$

70. C

Sol. ${}^nC_r : {}^nC_{r+1} : {}^nC_{r+2} \equiv 1 : 7 : 35$

$\Rightarrow \dfrac{r+1}{n-r} = \dfrac{1}{7} \Rightarrow 8r = n - 7$ and $\dfrac{r+2}{n-(1+r)} = \dfrac{1}{5}$

$\therefore\ n = 23$

SECTION – B

71. 7

Sol. $A^2 = \begin{bmatrix} 2^2 & 3^2 - 2^2 \\ 0 & 3^2 \end{bmatrix}$, $A^{10} = \begin{bmatrix} 2^{10} & 3^{10} - 2^{10} \\ 0 & 3^{10} \end{bmatrix}\ \therefore\ a + b + c + d = 2^1 \cdot 3^{10}$

$\therefore$ Number of factors $= 2 \times 11 = 22$

72. 5

Sol. Let r_1 and r_2 be the roots $r - 2 = \frac{1}{r} \Rightarrow r^2 - 2r - 1 = 0 \therefore b^2 + c^2 = 5$

SECTION – C

73. 16216.20

Sol. $\frac{^{12}C_4 \times {}^{15}C_4 \times \lfloor 4}{10^3} = 16216.20$

74. 00002.50

Sol. $|(x - 4) + i(y - 3)| = 2$; centre $\equiv (4, 3)$, radius = 2
$\therefore \alpha = 7$ and $\beta = 3$

75. 00000.50

Sol. $p = \frac{1}{2} \times \frac{4}{7} + \frac{1}{2} \times \frac{6}{8} = \frac{37}{56}$